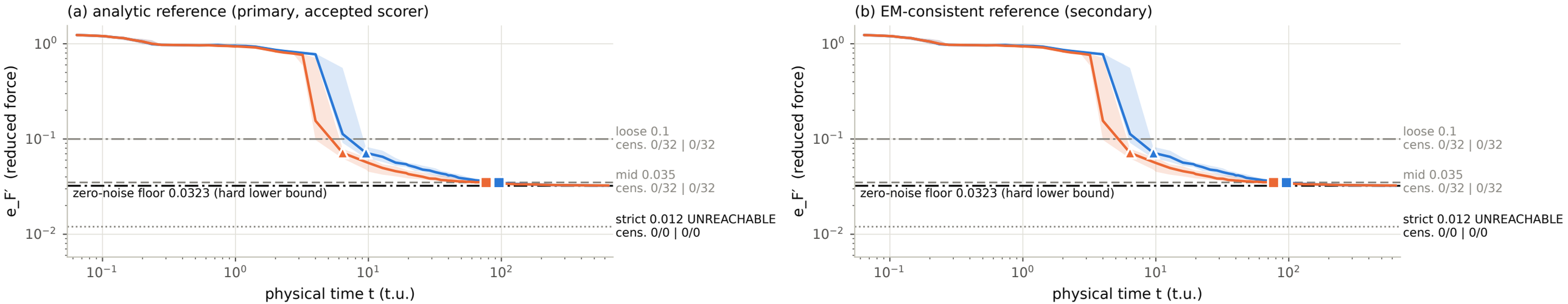
Gateway family expA/shift_lam16 · N = 128 · T_N = 640 t.u.

n_steps = 25,600,000 · h = 2.5e-05 · B = N n_steps = 3.277e+09 walker-steps · complete seeds: ABF 32/32, ABF+FR 32/32

Mean-force error e_F′ vs physical time



ABF (n = 32)
ABF+FR (n = 32)
interquartile range
strict threshold
mid threshold
loose threshold
EM bias of F′ at h (bin averages, analytic vs this dynamics' EM-consistent; NOT the e_F′ floor) -- numerically zero for this dynamics, not drawn
zero-noise floor (hard lower bound of the error)

Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.
Markers: first save after which the seed-MEDIAN curve stays at or below the threshold at every later save (persistent crossing of the median curve; the per-seed tau statistics are in summary.json). Censored thresholds have no marker; 'cens. a | b' next to each threshold = seeds whose own tau is censored (never persistently met by u = 1), ABF | ABF+FR.
Errors on the eval window [-1.5, 1.5] (151 grid nodes; e_F′ function space on the Gauss-Legendre sub-grid); Gamma = M / (C + 1).
e_F′ has a deterministic zero-noise floor 0.03227 (the within-bin variation of F′_ref, steep at the gate): e_F′² = RMS(Gamma − bin-averaged F′_ref)² + floor² for EVERY estimate, so a threshold below the floor is unreachable by construction (strict); 'both censored' there is not a tie. The shaded band is the EM bias of F′ (0), not the floor.